

TOPICAL REVIEW

Industrial Expert Systems Review: A Comprehensive Analysis of Typical Applications

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ABSTRACT As a branch of artificial intelligence (AI), expert systems are well-known for interpreting and deducing solutions to problems based on the rules contained within a knowledge base. Over the past decade, the application of expert systems in the industrial sector has matured significantly, permeating both upstream and downstream segments of the supply chain. Unlike traditional methods in industry, the approach using expert systems leverages the knowledge within domain experts, simulating the decision-making process of experts through rules and inference engines, rather than solely relying on the intuition of manual operators or engineers. In this study, the recent progress in the industrial application of expert systems were reviewed, including 1) the fundamental concepts of expert systems (e.g. the components of expert systems, expert systems based on different reasoning mechanisms, and the developmental history of expert systems) were briefly introduced; 2) the advantages and limitations of expert systems were discussed; 3) a series of detailed examinations for the expert system application in areas such as risk assessment, decision analysis, industrial classification, and fault diagnosis were elucidated in this study; 4) The challenges currently faced by expert systems and their future development prospects were discussed. This study not only reviewed the application of expert systems in the industrial field but also revealed how these systems drive automated decision support, reduce dependence on manual experience, and enhance operational efficiency and cost-effectiveness. Furthermore, the study highlighted future development directions. Overall, the research emphasized the importance of effectively integrating expert systems in industrial automation and intelligent manufacturing, providing valuable insights for practitioners and researchers.

INDEX TERMS Decision analysis, expert systems, fault diagnosis, industrial classification, risk assessment.

I. INTRODUCTION

The essence of expert systems is computer programs that simulate the knowledge and experience of domain experts to solve complex problems. They are capable of directly or indirectly acquiring knowledge from experts within a field, offering assistance to users lacking specific expert knowledge, and mimicking the decision-making process of human experts. By utilizing the expert knowledge stored within them, these systems provide users with precise solutions to their problems [1], [2], [3].

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The swift progression of industrial automation and intelligence has presented businesses with a growing number of complex decision-making obstacles. In the production process, quality control and assurance and production design and development can be difficult to manage without strong decision-making capabilities [4], [5]. Based on this context, the development of expert systems has gained significant prominence in integrating complex resources and data to enhance production efficiency in assembly line manufacturing. These systems are capable of processing large amounts of data and analyzing complex industrial processes to provide decision support based on deep knowledge and practical experience.

In the era of Industry 4.0, artificial intelligence (AI) technologies such as convolutional neural networks and support vector machines are driving intelligent manufacturing and digital transformation by analyzing real-time data to optimize production planning and predictive maintenance [6]. Expert systems can incorporate these AI technologies to offer more intelligent and efficient decision support. Furthermore, the application of asset management and condition monitoring technologies in industrial equipment is becoming increasingly important. Expert systems can optimize maintenance and management strategies through health indices and life-cycle management models, thereby enhancing reliability and extending equipment lifespan [7], [8].

Although the related research on the application of expert systems in industrial sector was mentioned in former literature, a comprehensive review of these research is still lacking. Therefore, a detailed review in the field of expert system applications should be proposed to analyze current research progress, identify existing gaps, and provide recommendations for future research directions to fill this important gap. To address this research gap, this study comprehensively reviews the application of expert systems in various industrial scenarios. For instance, in risk assessment, expert systems can conduct hazard analysis for chemical plant designs using the knowledge in their databases [9]; in decision analysis, expert systems can assist decision-makers in choosing the optimal option among multiple feasible plans by analyzing factors such as energy costs, system efficiency, and environmental impact in the application cases of combined heat and power systems [10]; in industrial classification, expert systems can accurately classify objects by analyzing ultrasonic sensor data, enhancing the level of automation and efficiency on production lines [11]; in fault diagnosis, expert systems can monitor the operational status of critical equipment like distillation columns in real time, predicting and diagnosing potential faults by analyzing data patterns, thereby reducing downtime and improving overall production efficiency [12].

This paper primarily reviews and analyzes the application of expert systems in the industrial field, including discuss the basic knowledge of various types of expert systems, the development history of expert systems, and a detailed discussion on how these systems are implemented in different industrial applications. The structure of the study is as follows: Section II will describe the basic knowledge of expert systems and provide a detailed introduction to three types of expert systems including Rule-based expert systems, Case-based expert systems and Neural network-based expert systems based on different reasoning rules, as well as the development history of expert systems. The current advantages and some limitations of expert systems will be discussed in Section III. Section IV will focus on the application of expert systems in the industrial field, including aspects such as risk assessment, decision analysis, industrial classification, and fault diagnosis. Section V will discuss some current challenges faced by expert systems and future development prospects. Finally, Section VI will offer suggestions to summarize the work of

this study. This study will provide a comprehensive overview of the application of industrial expert systems for researchers, engineers, and decision-makers in academia and industry. It will identify existing research gaps and potential future development directions, encouraging further exploration and utilization of the advanced capabilities of expert systems by academia and industry.

II. FUNDAMENTALS OF EXPERT SYSTEMS

Expert systems, as a potent artificial intelligence technology, are widely utilized in the industrial sector. Their ability to simulate the capabilities of human experts plays a vital role in industry.

As illustrated in Figure 1, the basic components of an expert system encompass the knowledge base, inference engine, user interface, interpreter, knowledge acquisition, and database [13], [14]. The knowledge base, serving as the system core, encapsulates the domain-specific knowledge and rules required for the expert system. This knowledge base is capable of storing and organizing knowledge in various representations, including rule-based, frame-based, and semantic networks. The inference engine works as the main actuator of the expert system, conducting inference and deduction based on the rules and facts within the knowledge base. It can generate pertinent conclusions by applying reasoning rules to the problems or scenarios presented by the users [15]. It may range from text-based command-line interfaces to graphical visual interfaces. Through this interface, users pose questions and receive generated conclusions or recommendations.

The interpreter is tasked with interpreting and parsing the inputs and questions from users, interacting with both the inference engine and the knowledge base. It is tasked with transforming user inputs into a format comprehensible to the inference engine, while also interpreting and conveying the engine's conclusions back to the users. Meanwhile, the knowledge acquisition system focuses on collecting insights from experts or external sources, subsequently organizing and storing this information in the knowledge base. This process may involve interviewing experts, analyzing documents, and simulating problem-solving processes; the database component focuses on the storage, retrieval, and management of data, emphasizing the organization and accessibility of information. It provides a mechanism for querying and updating data within the expert system, supporting the application of rules and the decision-making process within the knowledge base. This structured organization ensures the robustness and efficiency of the expert system functions, facilitating the handling of complex decision-making and problem-solving processes.

In addition to the fundamental components of expert systems, Figure 2 further delineates additional characteristics of expert systems. These include the benefits provided by expert systems (which will be elaborated in Section III), the supportive functionalities offered to operators in the industrial field, and the integrative interactions between expert systems and industrial automation equipment.

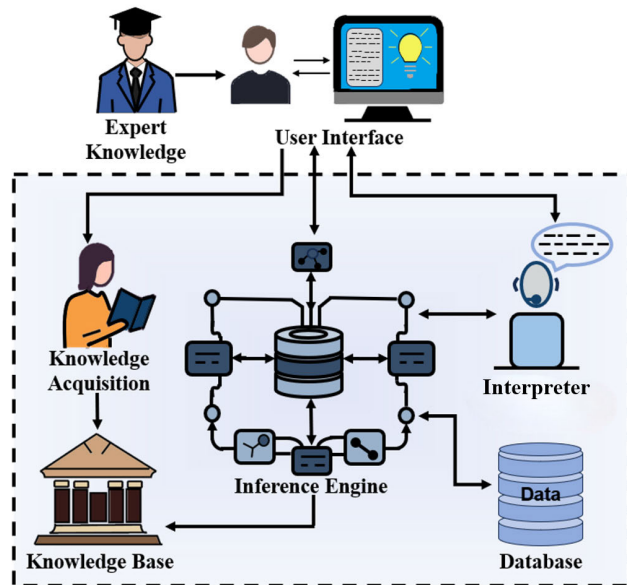


FIGURE 1. Components of an expert system.

A. EXPERT SYSTEMS BASED ON DIFFERENT REASONING MECHANISMS

Expert systems can be implemented based on various reasoning methods and are widely applied in the industrial field. Here, we introduce three types of expert systems in the industrial sector based on different reasoning mechanisms. The Table 1 below summarizes three types of expert systems used in the industrial sector: Rule-Based Expert Systems, Case-Based Expert Systems, and Neural Network-Based Expert Systems. For each type, it highlights their core ideas and functionalities, advantages, challenges, and modern enhancements. This comparative overview provides insights into how these systems mimic human decision-making, utilize historical cases, or learn from data, along with their respective strengths and areas needing improvement.

1) RULE-BASED EXPERT SYSTEMS

Rule-based expert systems saw significant development from the 1970s to the 1980s, marking an important direction in artificial intelligence research [16]. Initially designed to mimic the decision-making process of human experts in specific fields such as medicine, engineering, and finance, these systems primarily rely on “If-Then” rules to describe causal relationships and logical reasoning processes [17]. These expert systems use a predefined set of rules for reasoning and decision-making [18]. These rules outline the actions or inferences that should be taken in specific situations and are provided by domain experts based on their experience and knowledge. Since the rules directly represent the reasoning process of human experts rule-based expert systems are advantageous for their ease of understanding and explanation [18], [19]. While the addition and modification of a small number of rules are relatively straightforward, maintaining and updating the rule base can become increasingly complex

as the number of rules grows. Furthermore, these expert systems rely solely on predefined rules to handle problems, making it difficult to address new or exceptional situations not covered by the rules [20], [21], [22]. To address these challenges, many modern systems have started integrating machine learning techniques with traditional rule-based systems to enable a more dynamic decision-making process [16], [17].

2) CASE-BASED EXPERT SYSTEMS

Case-based expert systems began to gain attention in the 1980s. The core idea is to utilize historical cases to solve new problems, based on the notion that “past experiences are valuable resources for solving new problems” [23]. This approach is particularly suitable for solving problems that are difficult to address through strict algorithms. These expert systems identify solutions or suggestions by comparing the current problem with similar past cases, which contain information about the problem description, solution method, and outcome [19]. When encountering a new problem, the system searches for the most similar case and applies a similar solution method to generate recommendations. Case-based expert systems have learning capabilities, and as the number of accumulated cases increases, so does the system reasoning ability and accuracy. However, the performance of these systems is highly dependent on the quality and comprehensiveness of the case library. If the case library is not rich enough or is not updated in a timely manner, the quality of the reasoning may decline. Additionally, as the case library grows, the process of searching and comparing cases can become time-consuming, affecting the system’s response speed. To optimize case retrieval and management, more and more systems are adopting data mining techniques to automatically classify and update the case library, while using reinforcement learning to optimize the decision-making process.

3) NEURAL NETWORK-BASED EXPERT SYSTEMS

Neural network-based expert systems can trace their origins back to the 1940s, but it was not until the advancements in computer technology in the 1980s and 1990s that training complex networks became feasible, leading to the widespread application of neural networks in expert systems [24], [25]. These systems can learn and extract domain knowledge through training [4] and make inferences and decisions based on input data. These systems usually require a large amount of data for training to identify patterns and rules [26], [27]. They are capable of handling complex nonlinear relationships and possess adaptability and generalization capabilities. Neural network-based expert systems have advantages in processing image recognition, speech recognition, and big data analysis [28]. However, these systems typically require a large amount of labeled data for training, which can be challenging to obtain in some fields, and the “black box” nature of neural networks makes their decision-making process difficult to understand and

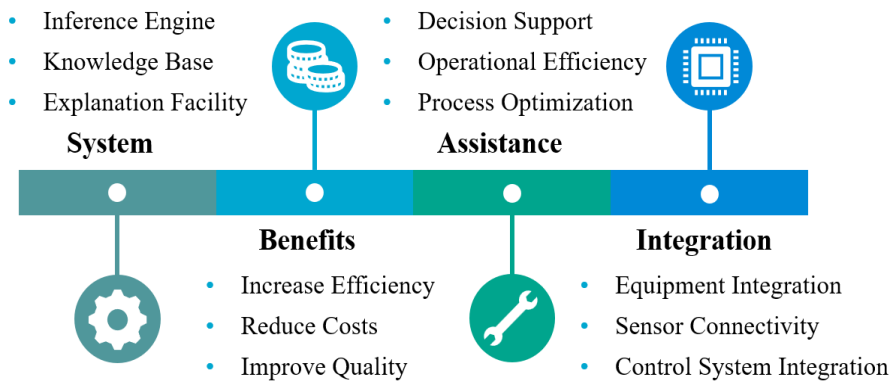


FIGURE 2. Characteristics of industrial expert systems.

TABLE 1. Expert systems based on different reasoning mechanisms.

Expert System Type	Core Idea and Functionality	Advantages	Challenges	Modern Enhancements
Rule-Based Expert Systems	Mimics decision-making process using "If-Then" rules, relies on predefined rules provided by domain experts to describe causal relationships and logical reasoning.	Easy to understand and explain, straightforward addition and modification of small number of rules.	Increasing complexity with the growth of rules, difficult to handle new or exceptional situations not covered by the rules.	Integrating machine learning techniques to enable dynamic decision-making.
Case-Based Expert Systems	Utilizes historical cases to solve new problems, based on the notion that "past experiences are valuable resources for solving new problems". Compares current problem with similar past cases to generate solutions or suggestions.	Learning capabilities improve reasoning ability and accuracy over time.	Performance dependent on the quality and comprehensiveness of the case library, time-consuming search and comparison process as the case library grows.	Adopting data mining techniques for automatic case classification and updates, using reinforcement learning to optimize decision-making.
Neural Network-Based Expert Systems	Learns and extracts domain knowledge through training, makes inferences and decisions based on input data, capable of handling complex nonlinear relationships, and possesses adaptability and generalization capabilities.	Handles complex tasks like image and speech recognition, big data analysis.	Requires a large amount of labeled data for training, "black box" nature makes decision-making process difficult to understand and explain.	Improving interpretability of neural networks to provide more insights into decision-making processes while maintaining performance.

explain [30], [39]. To overcome these issues, researchers are delving into the interpretability of neural networks, aiming to provide more insights into their decision-making processes while maintaining performance [31], [32].

In addition to the expert systems mentioned above, frame-based, fuzzy logic-based, and Dempster-Shafer theory based expert systems have also been proven applicable in the industrial sector. Each type has its unique advantages and applicable scenarios, and selecting the appropriate reasoning method for a specific requirement is crucial for the design and implementation of expert systems. In practice, the choice of expert system type depends on the nature of the problem, available data, real-time requirements of the system, and user needs. Sometimes, a hybrid approach that combines multiple types of expert systems can

leverage their advantages to better address the corresponding problems.

B. THE DEVELOPMENT HISTORY OF INDUSTRIAL EXPERT SYSTEMS

The year 1965 marks a significant milestone in the history of industrial expert systems. At that time, under the leadership of Feigenbaum at Stanford University in the United States, the development of the chemical expert system DENDRAL was initiated [33]. The DENDRAL system was dedicated to employing artificial intelligence technology to address issues related to chemical structures, achieving considerable breakthroughs in this domain. Table 2 summarizes the characteristics, representative examples, and application domains of various generations of industrial expert systems. From

TABLE 2. The development history of industrial expert systems.

Generation	Characteristics	Examples	Applications
First-Generation	Highly specialized, rule-based systems with strong problem-solving capabilities based on expert knowledge. Capable of reasoning and decision-making, offering valuable solutions for specific domain problems.	DENDRAL, MYCIN	Specific domain problems
Second-Generation	Single-discipline professional and application-type systems with advanced reasoning and decision-making capabilities. Utilize more sophisticated knowledge representation and reasoning technologies.	KADS, CYC	Medical diagnosis, engineering design, intelligence analysis
Third-Generation	Multidisciplinary integrated systems employing multiple AI languages and technologies. Not limited to single-discipline fields, offering flexible and powerful reasoning and decision-making capabilities. Utilize technologies like fuzzy logic, genetic algorithms, and neural networks.	Watson, AlphaGo	Multidisciplinary problems
Fourth-Generation	Use the latest AI technologies, including large-scale multi-expert collaborative systems and various knowledge representation methods. Focus on collaboration and knowledge sharing among experts to address complex, multidisciplinary problems. Employ multiple knowledge representation methods like ontologies and semantic networks, and integrated knowledge bases.	Google Cloud AI, Amazon AWS ML services	Complex, multidisciplinary problems, collaborative knowledge sharing

the highly specialized systems of the first generation to the fourth generation systems that utilize the latest artificial intelligence technologies and cloud computing services, the table illustrates the evolution of industrial expert systems from solving specific domain problems to addressing complex multidisciplinary issues. This progression highlights the significant enhancement in the scope and efficiency of industrial applications driven by technological advancements.

1) FIRST-GENERATION EXPERT SYSTEMS

First-generation expert systems, characterized by their specialization and strong problem-solving capabilities, include notable examples like DENDRAL [34] and MYCIN [35]. These systems are highly specialized, providing powerful solutions for specific domain problems based on rules and knowledge supplied. They are capable of reasoning and decision-making, offering valuable solutions to professionals.

2) SECOND-GENERATION EXPERT SYSTEMS

Second-generation expert systems are single-discipline professional and application-type systems [36], with KADS [37] and CYC [38] being typical examples. These systems are designed for professional applications in single-discipline fields, offering advanced reasoning and decision-making capabilities through more sophisticated knowledge representation and reasoning technologies. They are widely applied in fields like medical diagnosis, engineering design, and intelligence analysis.

3) THIRD-GENERATION EXPERT SYSTEMS

Third-generation expert systems are multidisciplinary integrated systems that employ multiple artificial intelligence languages [39], with Watson [40] and AlphaGo [41] as notable examples. These systems are not limited to single-discipline fields but possess multidisciplinary integrated characteristics. They use various artificial intelligence

languages and technologies, such as fuzzy logic, genetic algorithms, and neural networks, offering more flexible and powerful reasoning and decision-making capabilities.

4) FOURTH-GENERATION EXPERT SYSTEMS

Fourth-generation expert systems employ the latest artificial intelligence technologies, such as large-scale multi-expert collaborative systems, various knowledge representation methods, and integrated knowledge bases, like Google Cloud AI and various machine learning services of Amazon AWS. These systems are continuously evolving, focusing on collaboration and knowledge sharing among experts. By working collaboratively, they can address more complex and multidisciplinary problems. Moreover, fourth-generation expert systems adopt multiple knowledge representation methods, such as ontologies, semantic networks, and semantic reasoning, to better organize and manage knowledge. The use of integrated knowledge bases allows expert systems to acquire and integrate knowledge from different domain sources, providing more comprehensive and integrated solutions.

Over time, industrial expert systems have continuously evolved and developed, not only in terms of technological advancement but also in their evolution from specialized single-discipline systems to multidisciplinary and collaborative systems. This evolution reflects the constant adaptation and innovation of expert systems to the ever-changing and increasingly complex industrial environment, offering more powerful and flexible tools to address industrial challenges.

III. ADVANTAGES AND LIMITATIONS OF INDUSTRIAL EXPERT SYSTEMS

The application of expert systems in industry is becoming increasingly widespread, showcasing significant advantages in automation, intelligence, and efficiency. However, these systems also face with some challenges. To gain a comprehensive understanding of the performance of expert systems in the industrial field, Table 3 summarizes their main

TABLE 3. Advantages and limitations of industrial expert systems.

	Aspect	Description
Advantages of Industrial Expert Systems	Enhancement of Production Efficiency	Boosts efficiency by monitoring processes, identifying bottlenecks, and optimizing tasks and resources, reducing downtime and increasing output.
	Improvement of Product Quality	Ensures real-time quality control, detecting and correcting issues, reducing defects, and ensuring compliance with standards.
	Reduction of Labor Costs	Reduces the need for human resources by handling repetitive tasks, saving on training and management, allowing focus on strategic tasks.
	Enhancement of Reliability	Provides reliable decision-making based on strict rules, consistently performing tasks, reducing the risk of failures and disruptions.
	Facilitation of Knowledge Sharing and Transfer	Stores and shares expert knowledge across the organization, ensuring continuity and quality even after experts leave.
Limitations of Industrial Expert Systems	Dependence on Expert Knowledge	Relies on expert knowledge, which if outdated or insufficient, can compromise decisions, requiring frequent updates.
	Potential for Errors	Can make mistakes if rules are incomplete or inaccurate, necessitating regular checks and validation.
	Requirement of Significant Time and Resources	Developing and implementing these systems require considerable time and resources for knowledge engineering, software development, hardware procurement, and training.
	Need for Specialist Maintenance and Updates	Requires ongoing maintenance and updates, with specialists needed to update the knowledge base, monitor performance, and carry out repairs and improvements.
	Narrow Scope of Application	Generally focused on specific domains or tasks, limiting applicability to novel or intricate scenarios, requiring enhancements and multiple systems for diverse needs.

advantages and limitations. Expert systems offer notable benefits in enhancing production efficiency, improving product quality, reducing labor costs, increasing system reliability, and facilitating knowledge sharing. At the same time, challenges such as flexibility, maintenance costs, dependency on expert knowledge, and potential difficulties in handling complex situations should be emphasized. This table provides a clear overview of the specific advantages and limitations of expert systems in industrial applications.

A. ADVANTAGES OF INDUSTRIAL EXPERT SYSTEMS

Industrial expert systems offer numerous significant benefits in manufacturing and production, including enhanced productivity, better product quality, lower labor costs, reliability, and knowledge transfer.

1) ENHANCEMENT OF PRODUCTION EFFICIENCY

The primary advantage of industrial expert systems is the ability to increase production efficiency [42]. By monitoring the production process, equipment, and resource utilization, these systems can identify potential bottlenecks and optimization opportunities. They are capable of autonomously performing tasks, scheduling production, and allocating resources to ensure smooth operation of the production line [43]. This helps in reducing downtime while simultaneously increasing output and profitability [44].

2) IMPROVEMENT OF PRODUCT QUALITY

Industrial expert systems are pivotal for ensuring quality control in manufacture [45], which can monitor the production

process in real-time to detect and then correct quality issues promptly. With automated quality inspections and feedback, these systems can reduce the rate of product defects, ensuring compliance with quality standards and enhancing customer satisfaction.

3) REDUCTION OF LABOR COSTS

Implementing industrial expert systems can significantly reduce labor costs [46]. These systems can perform many repetitive and computation-intensive tasks, reducing the need for human resources [47]. This means companies can save time and money on staff training and management, enabling employees to focus on more creative and strategic tasks.

4) ENHANCEMENT OF RELIABILITY

Built on strict rules and logic, industrial expert systems offer highly reliable decision-making [37]. Unaffected by human operator emotions, they can consistently perform tasks under various conditions [48], enhancing system reliability and reducing the risk of accidental failures and production disruptions [49].

5) FACILITATION OF KNOWLEDGE SHARING AND TRANSFER

Expert systems are instrumental in the management of knowledge [45]. They can capture and store the knowledge of domain experts, making it available across the organization. This facilitates knowledge sharing and transfer, ensuring that valuable experience and expertise remain accessible even after experts leave or retire [50], which is critical for maintaining production continuity and quality consistency.

B. LIMITATIONS OF INDUSTRIAL EXPERT SYSTEMS

Despite their significant advantages in enhancing production efficiency and quality, industrial expert systems have some limitations and challenges that need to be considered during implementation and maintenance.

1) DEPENDENCE OF SYSTEMS ON EXPERT KNOWLEDGE

The performance of industrial expert systems depends on the knowledge previously provided by domain experts [49]. If the expert knowledge is insufficient or inaccurate, the system decision could be compromised [51]. Moreover, if experts are no longer available or their knowledge becomes outdated, the system might require significant modifications or rebuilding [52].

2) POTENTIAL OF SYSTEMS FOR ERRORS

Although designed to be highly reliable decision-making tools, industrial expert systems can still make mistakes [45]. If the rules in the knowledge base are incomplete or inaccurate, the system decision could lead to misleading results [48]. Therefore, the system performance must be regularly checked and validated to ensure its accuracy and reliability [53].

3) REQUIREMENT OF SIGNIFICANT TIME AND RESOURCES

Developing and implementing industrial expert systems typically requires considerable time and resources [54]. The knowledge engineering process involves acquiring, organizing, and structuring knowledge from domain experts, which is a complex task. Additionally, software development, hardware procurement, and staff training also entail significant costs and resources [55].

4) NEED OF SYSTEMS FOR SPECIALIST MAINTENANCE AND UPDATES

Industrial expert systems are not set-and-forget solutions; they require ongoing maintenance and updates. The knowledge base needs to be updated over time to reflect new technologies, processes, or market conditions [54]. Furthermore, specialists are needed to monitor the system performance and carry out necessary repairs and improvements [55].

5) SCOPE OF SYSTEMS' APPLICATION BEING NARROW

Industrial expert systems are generally engineered with a focus on particular domains or tasks, thereby constraining their applicability. Consequently, their capability to address novel or intricate scenarios may be restricted, necessitating ongoing enhancements and developments to cater to diverse requirements. In certain instances, the deployment of multiple expert systems might be imperative to encompass a wide spectrum of organizational demands.

IV. APPLICATIONS OF EXPERT SYSTEMS

As discussed in previous sections, this study focuses on research concerning the application of expert systems in

the industrial field. The industry still faces many challenges where the primary objectives of research are to enhance efficiency, reduce labor, and achieve automation in industrial production. Risk assessment in the industrial sector is regarded as a method to improve industrial safety. For instance, in the oil and gas industry, Zemenkov [56] developed a mathematical model for assessing and analyzing the reliability factors of pipeline transport and hydrocarbon storage facilities, which can effectively assess the risk to such facilities.

In the industrial domain, expert systems have been extensively employed due to the significant effectiveness in enhancing production efficiency, reducing costs, and ensuring product quality. The general development process of these expert systems, as illustrated in Figure 3, is divided into two critical phases: system design and modeling, and system implementation and optimization. During the design and modeling phase, the primary task is to clarify the specific domain of the problem, followed by the collection and articulation of expert knowledge, and the construction of a reasoning mechanism based on this knowledge. Upon completing this phase, the process transitions to the implementation and optimization phase, where the expert system is developed and integrated, followed by evaluation and debugging. If the evaluation results meet the standards of industrial applications, the development is considered successful; otherwise, the system requires further adjustments and optimizations, with subsequent evaluations until it meets the requirements. After development, it is essential to perform regular maintenance and upgrades on the expert system to maintain its efficacy, reliability, and security, ensuring its long-term effectiveness in the industrial production process.

This study mainly analyzes applications in the field of risk assessment [9], [52], [56], [57], [58], [59], [60], decision analysis [4], [5], [10], [50], industrial classification [11], [67], [68], [69], [70], [71], [72], [73], [74], fault diagnosis [12], [78], [79], [80], [81], [82], [83], [84], and other areas [87], [88], [89], [90], [91], [92], [93]. Furthermore, the Table 4 summarizes various application cases in these fields, highlighting the strong technical content and critical remarks. The technical content includes the operational processes in specific application scenarios, the algorithms and tools used, and data processing methods. The critical remarks provide an in-depth analysis of the performance and effectiveness of these technologies in real applications. These strong technical content and critical remarks not only demonstrate the superiority and practicality of each technology in real-world applications but also reveal their potential in different industrial environments.

Furthermore, Table 5 summarizes some examples applied in the aforementioned fields, highlighting specific areas such as power quality assessment, welding defect detection, and risk analysis, among others. This analysis underscores the diverse methodologies and technologies adopted in expert systems to enhance industrial processes, emphasizing the

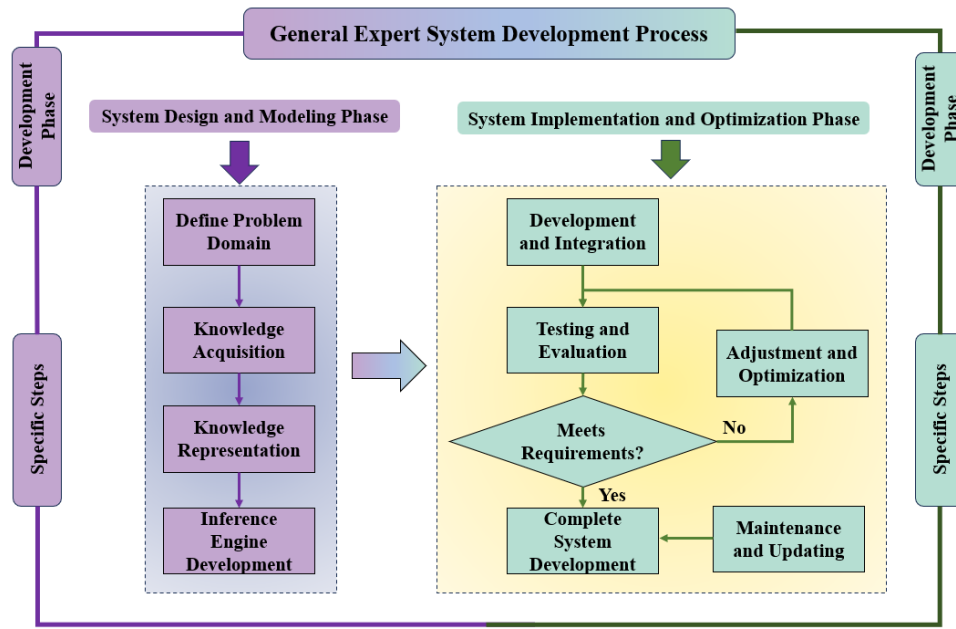


FIGURE 3. General expert system development process.

importance of continuous validation and improvement in real applications.

A. RISK ASSESSMENT

Many expert systems in the industrial sector have been applied to risk assessment with the aim of enhancing industrial safety, reducing production risks, improving efficiency, and minimizing factors that could lead to losses.

For instance, expert systems are widely used in hazard analysis for chemical plant design. Göring and Schecker [9] introduced a software tool named HAZEXPERT for this hazard analysis work. The core of HAZEXPERT is an object-based data model that can transform the graphical representation of CAD systems into dynamic object networks, followed by rule reasoning by a modularized general knowledge base. Additionally, Göring et al. developed a new systemic approach that combines the advantages of existing manual methods with the capability to structure and process large amounts of data, enhancing the integrity and efficiency of hazard analysis. This system also excels in asset maintenance, ensuring the safety and efficiency of plant operations through unified quality safety checks and documentation. The authors noted that the development of this system not only saves time but also avoids errors that may occur during manual data transfer, significantly improving the accuracy and reliability of the analysis.

Another significant application area of expert systems in risk assessment is the oil and gas industry. Brito and de Almeida [57] proposed a decision model based on multi-attribute utility theory, considering the impacts of pipeline incidents on personnel, environment, and economy and incorporating decision-makers' risk preferences and

behaviors for assessing the risk of natural gas pipelines. This model emphasizes the importance of asset maintenance in ensuring operational safety and reducing long-term costs by optimizing maintenance strategies and preventative measures. The authors commented on the contributions of this approach in enhancing the efficiency and precision of risk management, particularly its potential in predictive maintenance and reducing the incidence rate.

The reliability analysis of transportation and storage facilities for hydrocarbons is essential in the oil and gas industry. Zemenkov et al. [56] developed a mathematical model for assessing and analyzing reliability factors based on system analysis methods. These models consider the impact of production technology features, climatic factors, load factors, facility complexity, and maintenance characteristics on reliability, thus enabling risk assessment for oil and gas facilities. This study underscores the importance of quantitative assessment of these complex factors in improving facility maintenance and operational efficiency, particularly in preventative maintenance and risk management. The authors emphasized the efficacy of this approach, noting that it provides better support for the long-term operation of facilities, while making maintenance more scientific and precise through real-time data analysis.

More specifically, Tang et al. [58] proposed an integrated safety performance assessment framework, where each key safety factor corresponded to a list of basic safety factor indicators. Practitioners rated these indicators based on compliance, validating the standards of the safety performance evaluation framework and demonstrating the usability of 33 indicators, providing benchmarks and continuous improvement bases for the safety practices of Malaysian

TABLE 4. The strong technical content and critical remarks of application cases.

Research	Application Scenario	Strong Technical Content	Critical Remarks
Göring and Schecker [6]	Hazard analysis for chemical plant design	HAZEXPERT software tool with object-based data model and modularized knowledge base	Ensures safety and efficiency through unified quality safety checks and documentation, saving time and avoiding manual errors.
Brito et al. [42]	Risk assessment of natural gas pipelines	Multi-attribute utility theory, optimizing maintenance strategies and preventative measures	Emphasizes the importance of asset maintenance, improving risk management efficiency and precision, particularly in predictive maintenance and reducing incidence rates.
Zemenkov et al. [41]	Reliability analysis of oil and gas facilities	System analysis methods, real-time data analysis	Provides more scientific and precise maintenance support, enhancing long-term operation and risk management, especially in preventative maintenance.
Tang et al. [43]	Safety performance assessment of Malaysian offshore oil and gas platforms	Safety performance evaluation framework, 33 safety performance indicators	Ensures the normal operation of equipment and systems, reducing accident risks and improving overall safety management efficiency, with the ability to predict future safety risks and timely adjustments.
Wang et al. [44]	Prediction of chemical reactions between drugs and excipients	PharmDE expert system, compatibility risk assessment based on structural features	Supports drug formulation design, reducing incompatibility risks, improving risk management efficiency, and supporting more rational experimental design.
Shen et al. [45]	Risk assessment of cold chain supply chain	Expert system knowledge base and entropy-OWA method, real-time management	Enhances the efficiency and value of the cold chain supply chain, particularly in food safety and reducing transportation losses, proving the effectiveness and practicality of the comprehensive assessment method.
Ngan et al. [37]	Prioritizing risk mitigation strategies for biomass polygeneration systems	Decision and Evaluation-based Fuzzy Analytic Network Process (DEFANP) method, Monte Carlo simulation	Ensures the long-term stability and production efficiency of biomass polygeneration systems by selecting strategies with the highest analytical indicators and coordinating management objectives.
Piltan et al. [7]	Identifying and prioritizing industries for CHP systems in Iran	Decision model combining AHP and FES, considering seven criteria and ranking 23 industries based on ISIC	Emphasizes the critical role of asset maintenance in reducing energy consumption and enhancing efficiency, demonstrating significant improvements in asset energy efficiency and economic returns.
Ali et al. [5]	Natural composite fibers as alternatives to synthetic materials in the automotive industry	AHP method with a four-level hierarchical model, including physical and mechanical properties, environmental factors, and costs	Reinforces environmental responsibility awareness, enhances asset maintenance capabilities, and improves environmental sustainability and resource utilization efficiency.
Rahman et al. [4]	Problem-solving in the automotive repair industry	CBR cycle (retrieving, reusing, revising, retaining) and OOSE model for online application	Improves efficiency and quality of asset maintenance by integrating past experiences, reducing vehicle failure response time and repair costs, and strengthening fault diagnosis accuracy.
Merten et al. [35]	Decision-making strategies in steel manufacturing	Network analysis of human expert decision strategies, representing strategies through nodes and dependency-based networks	Highlights asset maintenance as crucial for maintaining production continuity and reducing risks, utilizing historical production data to reveal decision-making complexity and identify pivotal nodes.
Llata et al. [8]	Object classification using ultrasonic sensors	Dual knowledge base fuzzy expert system combining statistical and rule knowledge	Enhances operational accuracy and reliability by adapting to environmental changes and uncertainties, improving production efficiency and reducing failure rates.
Liao et al. [46]	Classification of welding defects	Fuzzy expert system generating and comparing fuzzy rules, evaluated against other methods	Improves maintenance efficiency and reduces the risk of misdiagnosis, standardizing and automating welding quality control, significantly enhancing asset reliability and production efficiency.
Rearden et al. [47]	Fuel classification	GC/DMS combined with a fuzzy rule generation expert system (FuRES)	Provides reliable fuel identification and classification, ensuring fuel quality assurance and asset maintenance, with high efficiency and accuracy.
Sun et al. [48]	Jet fuel classification	Expert system (FuRES) for fast GC-fast MS data classification	Enhances efficiency of asset maintenance management, providing robust technical support for aviation fuel safety through precise and rapid data analysis, especially in real-time monitoring.
Patel et al. [49]	Ore classification in mining	SFFS algorithm for feature selection and SVM for classification model	Optimizes feature selection and improves classification accuracy, enhancing ore processing efficiency and quality control, reducing reliance on human resources, and minimizing errors.

TABLE 4. (Continued.) The strong technical content and critical remarks of application cases.

Brüangel et al. [50]	Classification of nanomaterials	NanoDefiner e-Tool for nanomaterial classification	Promotes standardization and normalization of nanomaterials, ensuring correct classification, protecting investments, and ensuring product quality.
Uyar et al. [51]	Power quality disturbance classification	Expert system based on S-transform and neural networks	Helps maintenance personnel take timely measures by real-time monitoring and automatic identification of power quality disturbances, improving asset management efficiency and operational safety.
Rajeshbabu and Manikandan [52]	Power quality disturbance classification	LDA for feature extraction, AHP for knowledge base, GMDH and ANFIS for classification	Improves accuracy of diagnosing and classifying power quality problems, crucial for maintaining health and functionality of power grid assets, optimizing maintenance strategies and reducing repair costs.
Jin and Chen [53]	Vibration signal classification	End-to-end framework with data preprocessing, feature extraction, modified Transformer network, and overall optimization	Significantly improves efficiency of operational monitoring and maintenance of mechanical structures, promptly detecting and preventing potential faults, reducing downtime and maintenance costs.
Leung and Romagnoli [9]	Fault diagnosis of a pilot-scale distillation column	Probabilistic model-based fault diagnosis expert system	Significantly improves operational efficiency and reliability through real-time monitoring and precise diagnosis, optimizing maintenance strategies and minimizing unplanned downtime.
Ma et al. [54]	Fault diagnosis of transformers and high-voltage vacuum circuit breakers	Multi-layer BP network method	Enhances diagnostic efficiency and accuracy, optimizing maintenance strategies, increasing system reliability and stability, and achieving predictive maintenance and fault prevention.
Purkait and Chakravorti [55]	Fault diagnosis of transformers	Time-frequency analysis-based fault diagnosis expert system	Greatly improves maintenance and management efficiency through precise fault identification and location, enhancing fault diagnosis speed and accuracy, and reducing downtime.
Burriel-Valencia et al. [56]	Fault diagnosis of induction motors (IMs)	Hyper-system implementation for fault diagnosis	Improves asset maintenance efficiency through real-time accurate monitoring and fault analysis, optimizing and streamlining the fault diagnosis process, and reducing maintenance costs.
Qian et al. [57]	Fault diagnosis of the FCCU in petrochemical processes	Real-time fault diagnosis system	Significantly improves asset operational efficiency and reliability through real-time monitoring and diagnosis, preventing potential catastrophic accidents and ensuring equipment safety and production stability.
Qian et al. [58]	Improved fault diagnosis of the FCCU	Real-time expert system based on wavelet transform and fuzzy ART neural network	Enhances system capabilities in real-time monitoring and fault prediction, optimizing maintenance strategies, reducing equipment failure rates, and extending equipment life.
Jie and Shen [59]	Fault diagnosis of aviation electronics equipment	Knowledge base architecture based on Fault Query Network (FQN)	Provides comprehensive and accurate fault diagnosis support, enhancing maintenance and reliability of critical aviation assets, reducing misdiagnosis rates, and improving maintenance decision quality.
Niccolai et al. [60]	Automated detection in aerospace production lines	Vision-based expert system for defect detection using image recognition technology	Significantly improves maintenance efficiency by enhancing detection speed and accuracy, reducing human error, and ensuring continuity of production processes and product quality.

offshore oil and gas platforms. In this framework, the importance of asset maintenance is explicitly emphasized, as it directly enhances the overall safety performance of the platform by ensuring the normal operation of equipment and systems, thereby reducing the risk of accidents. This method not only assesses current safety performance but also predicts future safety risks, thus aiding in timely adjustments to safety measures and improving overall safety management efficiency.

Another promising field of risk assessment is pharmaceuticals and biotechnology. Wang et al. [59] developed

an expert system named PharmDE that predicted potential chemical reactions between drugs and excipients based on their structural features, assessing compatibility risks and reducing potential risks during the process. The significance of this system lies in its support for drug formulation design by reducing the risks associated with incompatibility with excipients, optimizing asset maintenance management. The authors believe that this system not only improves the efficiency of risk management but also supports more rational experimental design, thereby protecting the long-term sustainability of assets.

TABLE 5. Analysis of the current status of expert systems applications in the industrial field.

Citation	Application	Main Features	Validation
Abdelsalam et al. [101]	Power Quality Assessment and Diagnosis	Fourier Linear Combiners, Minimum Mean Square Error Criterion	Simulation Software, Experimental Verification
Liao [67]	Welding Defect Detection	Optimal Fuzzy Partitioning SGA-WM Method, Bootstrap Method	Testing with Real Industrial Data Sets
Ceylan [102]	Risk Analysis of Shipboard Compressed Air Systems	Rule-Based Fuzzy Failure Mode and Effects Analysis (FMEA) Method	None
Sun et al. [69]	Classification and Identification of Jet Fuels	Fast Gas Chromatography - Fast Scan Quadrupole Ion Trap Mass Spectrometry	Real Experimental Verification
Ma et al. [78]	Power System Fault Diagnosis	Multi-Layer Perceptron Neural Networks	Validation with Real Power System Data Sets
Wu et al. [103]	Detection and Classification of Different Working States and Fault Types in Internal Combustion Engines	Wavelet Packet Transform, Artificial Neural Networks	Experiments with Gasoline Direct Injection (GDI) Engines
Zhi-Ling et al. [104]	Wind Turbine Gearbox Fault Diagnosis	Fault Tree Analysis	None
Niu and Zhao [105]	High Voltage Circuit Breaker Fault Diagnosis	Radial Basis Function Neural Network, Explanation and Reasoning Mechanisms	Validation with Switching Current Curve Samples
Yang et al. [106]	Vibration Fault Diagnosis of Rotating Machinery	Causal Matrices, Bayesian Algorithms	Verification with a Centrifugal Pump Vibration Fault Case
Chan [107]	Analysis of Oil Production and Separation Processes	Knowledge Modeling System, Sensor Data Analysis	None
Lu et al. [108]	Fault Diagnosis of Reciprocating Compressors	Web-Based Intelligent Diagnostic Procedure	None
Trivedi et al. [109]	Tensile Strength Prediction of FFF Manufactured Parts	Fuzzy Logic	Real Data Validation
Adamczyk et al. [110]	Semantic Interoperability and Knowledge Sharing in Manufacturing Systems	Ontologies as Knowledge Bases, Reasoning Engines for Semantic Interoperability and Knowledge Retrieval	Validation in an Automotive Manufacturing Industry

TABLE 5. (Continued.) Analysis of the current status of expert systems applications in the industrial field.

Hassannayebi et al. [111]	Fault Diagnosis of Natural Gas Flow Control Facilities	Decision Trees, Dynamic Control Flow Charts, Design with C# Programming Language	Real Experimental Validation
Ghanei et al. [112]	Online Characterization of Microstructures in Dual-Phase Steel	Barkhausen Noise Testing Technique, Adaptive Neuro-Fuzzy Inference System	Validation with Real Data
Žarković and Stojković [113]	Condition Monitoring and Fault Diagnosis of Electric Power Transformers	Fuzzy Logic, Various Online and Offline Transformer Testing Methods	Testing with Four Different Voltage Level Transformers
Rodriguez-Alabanda et al. [114]	Optimization of Multi-Pass Metal Drawing	Slab Method, Finite Element Analysis Software Deform2D for Simulation and Verification	Real Experimental Verification
Mitiche et al. [115]	Condition Monitoring and Fault Diagnosis of High Voltage Equipment	Bispectrum, Convolutional Neural Networks	Validation with Real-World EMI Signal Data Sets
Berredjem and Benidir [116]	Bearing Fault Diagnosis	Overlapping Range Method, Fuzzy Inference Engine	Validation with Public Bearing Fault Data Sets
Cebi et al. [117]	Ship Machinery Fault Detection	Backward Linkage Reasoning Engine	None
Chen et al. [118]	Safety System Design of Autonomous Underwater Vehicles	Knowledge Bases, Reasoning Engines	Simulation Experiments
Dash et al. [119]	Classification of Transient Disturbance Waveforms in Power Systems	Fourier Linear Combiners	Validation with Simulated Data
Hajnayeb et al. [120]	Fault Detection and Diagnosis of Rolling Element Bearings	UTA Algorithm, Multilayer Perceptron	Validation with Typical Rolling Element Bearing Experimental Data Sets
Liang et al. [121]	Fault Diagnosis of Shipboard Electrical Power Systems	Knowledge Bases, Reasoning Mechanisms, Fuzzy Rules	None
Nan et al. [122]	Real-Time Fault Diagnosis in Chemical Processes	Fuzzy Inference System	Validation with Two Types of Cases
Xu et al. [123]	Fault Diagnosis of Electric Metering Devices	Forward Reasoning Mechanism, Fault Knowledge Base	None
Cobos-Guzman et al. [124]	3D Printing Technology	Fuzzy Logic, Taguchi Method	Simulation Software Testing

TABLE 5. (Continued.) Analysis of the current status of expert systems applications in the industrial field.

Hu et al. [125]	Rail Circuit Fault Prediction	Grey Theory	Fault Prediction for High Voltage Asymmetric Pulse Rail Circuit
Tuitemwong and Premrudee preechacharn [126]	Protection Coordination in Distribution Systems with Distributed Generation	Knowledge Bases and Reasoning Process to Optimize Protection Devices	Testing with a 22 kV Distribution System
Simonova et al. [127]	Technical Preparation and Technological Route Selection in Production Processes	Frame Representation, Integration with Multi-Agent Systems	None
Merino et al. [128]	Suppression of Cutting Vibrations During Train Axle Repair	Embedded Sensors, Cloud Computing	Industrial Experiment Verification
Jong et al. [129]	Multi-Level Workshop Scheduling in Mold Manufacturing	GA and ACO Metaheuristic Algorithms	Experimental Validation with Four Real Mold Cases
Wang et al. [130]	Maintenance of Pressure Safety Systems in Nuclear Power Plants	Multi-Layer Flow Model, Backward Linkage	Validation with a Pressurized Water Reactor Simulator
Ramezani and Jassbi [131]	Improvement of Quality and Efficiency in Production Processes	Statistical Process Control, Learning Vector Quantization Network and Seven Multilayer Perceptron Networks	Learning vector quantization network and seven multilayer perceptron network
Dong et al. [132]	Automatic Control of Coal Slime Flotation Process	Least Squares Support Vector Machine, Fuzzy Control	Validation with a Real Coal Slime Flotation Column
Aguilera et al. [133]	Control of Robotic Arm Movement	Inverse Kinematics Model, Cubic Polynomial Trajectory, Blockchain Technology and IoT Platform	Validation with an Experimental Robotic Arm with Degrees of Freedom
Sandak et al. [134]	Quality Control of Wood and Wood Products	Portable NIR Spectrophotometer, PLS-DA, kNN, SIMCA, SVM Methods	Real Specific Experiments Validation
Du et al. [135]	Human-Robot Collaboration	Kinect Sensor Monitoring, Unscented Kalman Filter, Artificial Potential Fields	Validation with Robots and Kinect Sensors
Muślewski et al. [136]	Optimization of Operation and Maintenance Processes in Complex Technical Systems	Multi-Objective Analysis, Semi-Markov Process	Simulation Software for Authentication

TABLE 5. (Continued.) Analysis of the current status of expert systems applications in the industrial field.

Sánchez et al. [137]	Risk Assessment of Exposure to Hand-Arm Vibration	Risk Matrix, Fuzzy Logic	Simulation and Real Case Validation
Nourian et al. [138]	Natural Gas Pressure Reduction Stations	Fuzzy Logic, Knowledge Extraction and Abstraction	Validation in a Real Plant

The cold chain supply chain is a complex process of processing, storing, transporting, and distributing goods in a low-temperature environment, involving multiple enterprises and links, facing risks from refrigeration, supply and demand, logistics, information, and environment. To address these risks, Shen and Qian [60] proposed an assessment method using the expert system knowledge base and entropy-OWA method, combining the knowledge and experience of expert groups to assign values to risk indicators and adjusting expert weights to reduce subjective errors. Then, the adjusted risk values are effectively evaluated through the OWA operator, ranking the risk values, and implementing refined management, intelligent management, real-time management, and risk avoidance management measures for the cold chain supply chain to ensure food quality safety, improve the efficiency and value of the cold chain supply chain. The authors also emphasized the importance of this system in asset maintenance, especially in improving food safety and reducing losses during transportation, proving the effectiveness and practicality of this comprehensive assessment method.

In biomass polygeneration systems, prioritizing risk mitigation strategies is crucial as these measures directly relate to the sustained operation and maximization of benefits. Ngan et al. [52] proposed a Decision and Evaluation-based Fuzzy Analytic Network Process (DEFANP) method combining analytical models and stochastic programming to evaluate and select risk mitigation strategies for biomass polygeneration systems, prioritizing risk mitigation strategies among the project objectives, key components, and industrial stakeholder networks of the biomass industry. Figure 4 illustrates these network relationships, highlighting the interactions between different elements and groups, thus providing valuable insights for understanding and determining the priorities of asset maintenance and risk mitigation strategies. Monte Carlo simulation methods can simulate the fluctuations and uncertainties in operation, transportation, market supply and demand, and prices. By selecting strategies that produce the highest analytical indicators and coordinate with the corresponding probabilities of achieving management objectives, the risks of implementing biomass polygeneration systems can be mitigated, ensuring the system’s long-term stability and production efficiency.

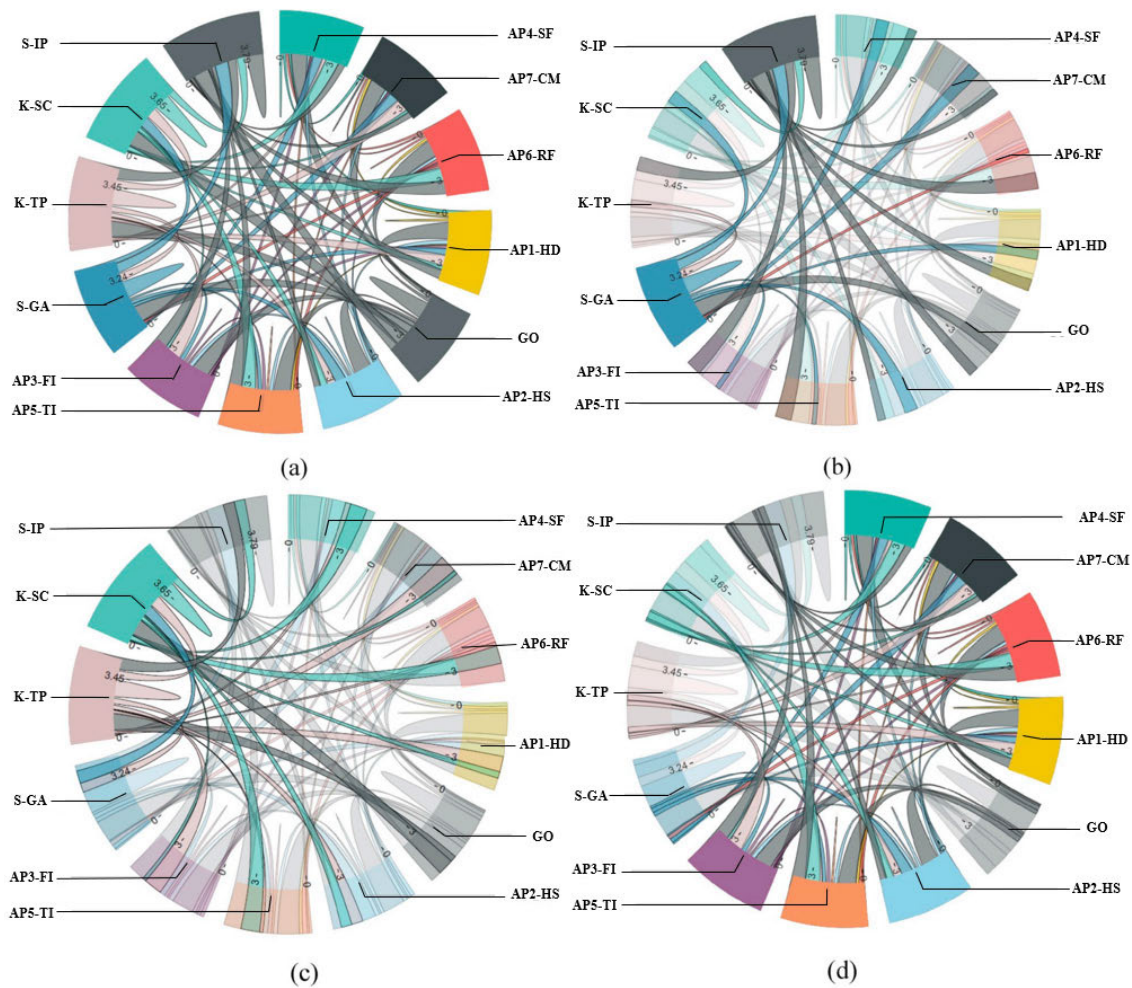


FIGURE 4. DEFANP relationship nexus highlighted for: (a) overall elements; (b) stakeholder cluster; (c) key elements cluster; (d) action plan cluster [52].

Yan et al. [61] developed a new risk assessment model using expert systems to improve the Risk Priority Number (RPN) calculation in Failure Modes and Effects Analysis (FMEA). They emphasized the importance of this model in asset maintenance, particularly in managing risks under complex and uncertain environments.

In the area of energy consumption decision analysis for electric vehicles, Deptuła et al. [62] developed an expert system using Multi-Valued Logic Trees to evaluate the energy consumption of electric vehicles under different driving cycles. This system has significant applications in asset maintenance. By simulating the energy consumption of electric vehicles under various ambient temperatures and average speed conditions, it provides decision support for drivers.

In the health condition diagnosis of lithium batteries, Xiao et al. [63] developed an expert system for the intelligent diagnosis of the health condition of lithium battery electrode roll pressing equipment. This system has significant applications in asset maintenance. By integrating an improved BP neural network and fuzzy logic, the system effectively

links neural networks with health assessment to achieve a comprehensive evaluation of the equipment's operational state.

B. DECISION ANALYSIS

In the field of decision analysis, expert systems continue to play an irreplaceable role such as production optimization [50] and energy management [5]. Expert systems provide highly intelligent support in industrial decision analysis, helping to increase production efficiency, reduce risks, and improve decision quality, thereby bringing benefits to organizations in multiple aspects.

To identify and prioritize potential industries for installing Combined Heat and Power (CHP) systems in Iran, Piltan et al. [10] proposed a decision model combining the Analytical Hierarchy Process (AHP) and Fuzzy Expert System (FES). The model considered seven influencing factors or criteria, used AHP to calculate the weight of each criterion, and ranked 23 Iranian industries according to the

International Standard Industrial Classification (ISIC) using FES based on real data. The results showed that industries such as chemicals, non-metallic minerals, refined petroleum, food and beverages, basic metals, paper, and textiles are more suitable for installing CHP systems than other industries. This study emphasized the critical role of asset maintenance in reducing energy consumption and enhancing efficiency, demonstrating that effective deployment of CHP systems significantly improves asset energy efficiency and economic returns. Additionally, the authors highlighted the potential applications of their model in energy policy formulation and optimization of industrial energy structures.

Similarly, when exploring natural composite fibers as an alternative to synthetic materials in the automotive industry, Ali et al. [5] also used the AHP method to establish a four-level hierarchical model, including objectives, criteria, sub-criteria, and candidate materials. Criteria included physical properties, mechanical properties, environmental factors, and costs; sub-criteria included density, tensile strength, Young's modulus, biodegradability, toxicity level, raw material cost, and manufacturing cost; candidate materials included polypropylene (PP) and seven different natural fiber-reinforced polypropylene composites (PP/Kenaf, PP/Sisal, PP/Flax, PP/Hemp, PP/Jute, PP/Coir, PP/EFB). Pairwise comparisons were conducted using the Expert Choice software tool to calculate the weight vector and consistency ratio and perform sensitivity analysis. This method not only reinforced awareness of environmental responsibility within the automotive industry but also enhanced the asset maintenance capabilities through an intelligent selection system. The authors thoroughly evaluated the effectiveness of this decision support system, highlighting its significant importance in improving environmental sustainability and optimizing resource utilization efficiency.

The conclusion proved that the AHP method was very effective in solving multi-criteria decision-making problems, and the results were validated through consistency ratio and sensitivity analysis. Moreover, the selected biodegradable materials also reduced the use of petroleum-based materials, thereby enhancing passenger safety.

Additionally, in the automotive repair industry, Rahman et al. [4] proposed a new approach to problem-solving based on past similar cases, including the four main stages of the CBR cycle (i.e., retrieving similar problems, reusing knowledge, revising solutions, and retaining experience). Furthermore, they used the Object-Oriented Software Engineering (OOSE) model to build an online application for specific application, providing great convenience for car users and technicians. This approach significantly improved the efficiency and quality of asset maintenance by integrating past experiences and cases, reducing vehicle failure response time and repair costs. The case-based reasoning method not only enhanced the accuracy of fault diagnosis but also strengthened the systematic and scientific nature of maintenance decisions, having a significant impact on technological advancements

and knowledge management in the automotive repair industry.

In the steel manufacturing industry, Merten et al. [50] explored the factors and dependencies in decision-making strategies through network analysis of human expert decision strategies. Using a network analysis framework, the study represented expert decision strategies through nodes and constructed dependency-based networks. The investigation into network analysis encompassed aspects such as node interrelations, the overall network structure, the significance of individual nodes, and the direction of information flow. In this process, asset maintenance, as a crucial component of decision support, was highlighted for its key role in maintaining production continuity and reducing risks. Through this approach, historical production data can be effectively utilized to reveal the complexity of production decisions, pivotal nodes and factors influencing the decision-making process were identified, leading to recommendations for enhancements.

Güneri and Deveci [64] explored a novel fuzzy decision-based framework for supplier selection in the defense industry. This framework has significant applications in asset maintenance. The framework integrates various expert system techniques and methodologies, including the Delphi method, Analytic Hierarchy Process (AHP), and a fuzzy set method known as q-Rung Orthopair Fuzzy Sets (q-ROFS) based on the Evaluation based on Distance from Average Solution (EDAS) method. In this way, the framework provides more effective decision support in industrial environments.

Wang et al. [65] developed an expert system that combines signal fluctuation processing techniques with a collaborative filtering recommendation algorithm to optimize and enhance the accuracy of recommendation systems. This system has significant applications in asset maintenance. By analyzing the volatility and instability of user ratings, they introduced a new method called the kurtosis factor to assess and determine the credibility of expert users. The system specifically addresses the issue of expert ratings in traditional collaborative filtering algorithms not being properly screened.

Akram et al. [66] developed a new method called "Hesitant Fuzzy Soft Expert Sets" (HFSE sets). This method has significant applications in asset maintenance. By combining the characteristics of Hesitant Fuzzy Sets and Soft Expert Sets, this approach allows decision-makers to more effectively express and process their opinions and information in uncertain and ambiguous environments. The study presents a practical application case of selecting an electric vehicle model, demonstrating the actual application and effectiveness of this model, highlighting its advantages over existing models.

C. INDUSTRIAL CLASSIFICATION

In traditional industrial classification, completing a classification task demands considerable time and human resources.,

making the process extremely inefficient. Therefore, when carrying out industrial classification tasks, the imperative to enhance classification efficiency and achieve swift, precise categorization has become the most pressing issue in this field.

Llata et al. [11] developed a method for object classification using ultrasonic sensors and a dual knowledge base fuzzy expert system. This method used a pair of ultrasonic sensors to obtain information about the shape of an object from the envelope of the ultrasonic echo signal, thereby extracting features of the object shape for classification. The feature of this method was the combination of statistical knowledge (knowledge extracted from data) and rule knowledge (human expert knowledge) in a reasoning engine, using the normal distribution function as the membership function to automatically generate simple and repetitive rules, capable of handling uncertainties and environmental changes during ultrasonic propagation. This system is specifically designed for industrial applications, where asset maintenance is a core consideration. By intelligently adapting to environmental changes and uncertainties, the system enhances operational accuracy and reliability. The authors highly praised the efficiency and adaptability of this system, emphasizing its contribution to improving production efficiency and reducing failure rates in practical applications.

In the classification of welding defects, Liao [67] proposed a method based on a fuzzy expert system. In their method, fuzzy rules were generated by two different methods from existing instances, and the fuzzy rules generated by the two methods were evaluated and compared for their classification accuracy in the fuzzy expert system. Furthermore, the authors compared the fuzzy expert system method with the bootstrap-based fuzzy nearest neighbor algorithm and multilayer perceptron neural network. In the asset maintenance of the welding process, this method significantly enhances asset reliability and production efficiency by improving the accuracy and consistency of defect identification. The results showed that the fuzzy expert system method outperformed other methods. The authors highlighted the advantages of the fuzzy expert system in improving maintenance efficiency and reducing the risk of misdiagnosis, which is of great significance for standardizing and automating welding quality control.

In the field of fuel classification, Rearden et al. [68] proposed a method using gas chromatography/differential mobility spectrometer (GC/DMS) combined with a fuzzy rule generation expert system (FuRES) to develop an expert system capable of classifying and identifying 12 different fuels. This system is crucial for fuel quality assurance and asset maintenance, particularly in providing reliable fuel identification and classification. The authors also explained the efficiency and accuracy of this method, emphasizing its stability and repeatability in practical applications.

Differently, Sun et al. [69] constructed a fuzzy rule generation expert system (FuRES) to classify fast gas chromatography-fast scan quadrupole ion trap mass

spectrometry (fast GC-fast MS) data for jet fuel classification. This greatly enhances the efficiency of asset maintenance management, timely detecting subtle changes in fuel to ensure its quality and performance. The application of this system in jet fuel monitoring shows significant potential, providing robust technical support for aviation fuel safety through precise and rapid data analysis, especially in real-time monitoring and asset maintenance.

Both methods achieved satisfactory results in the field of material classification. Compared with traditional classification methods, using expert systems for material classification can complete tasks more rapidly and accurately.

In ore collection, Patel et al. [70] developed an expert system using the Sequential Forward Floating Selection (SFFS) algorithm to select the optimal feature subset and the Support Vector Machine (SVM) algorithm to develop a classification model. This expert system can quickly and accurately classify iron ore, providing the mining industry with a fast and inexpensive method to replace traditional human visual monitoring systems for ore classification. The system's application in asset maintenance is significant, optimizing feature selection and improving classification accuracy, greatly enhancing ore processing efficiency and quality control, and reducing reliance on human resources. This method not only improves operational efficiency but also reduces maintenance costs by minimizing errors and enhancing prediction accuracy, contributing significantly to the sustainable development of the mining industry.

Additionally, in the classification of nanomaterials, Brüngel et al. [71] developed an expert system called NanoDefiner e-Tool. NanoDefiner e-Tool can provide valuable support for nanomaterial manufacturers, measurement technology developers, and independent analytical laboratories, promoting the standardization and normalization of nanomaterials. The system particularly emphasizes the maintenance of nanomaterial assets, providing a scientific basis for the correct classification of nanomaterials, thus protecting investments and ensuring product quality. The authors positively evaluated the practicality and promotional value of this tool in their paper, highlighting its important role in supporting the development of nanotechnology.

In the field of power quality disturbance classification, Uyar et al. [72] developed an expert system based on S-transform and neural networks for the automatic classification of power quality disturbances. The system considered nine power quality disturbance categories, including normal voltage, voltage sag, voltage swell, short interruption, flicker, oscillatory transient, harmonics, sag with harmonics, and swell with harmonics, and can effectively classify different power quality disturbances with high accuracy and robustness. In asset maintenance, this system can help maintenance personnel take timely measures by real-time monitoring of grid quality and automatic identification of disturbance events, thus improving asset management efficiency and operational safety. Furthermore, Uyar et al. pointed out that this system can provide important support for the reliability

and stability of the power system, emphasizing the need to further optimize the algorithm and expand system functions to meet the challenges of more complex scenarios in the future.

Subsequently, in the field of power quality disturbance, Rajeshbabu and Manikandan [73] used Linear Discriminant Analysis (LDA) for feature extraction from raw data, then used the Analytical Hierarchy Process (AHP) to construct the knowledge base of the expert system, and finally used the Group Method of Data Handling (GMDH) and Adaptive Neuro-Fuzzy Inference System (ANFIS) to classify power quality disturbances. This comprehensive method not only improves the accuracy of diagnosing and classifying power quality problems but is also crucial for maintaining the health and functionality of power grid assets, helping system operators optimize maintenance strategies and reduce repair costs. In practical applications, this method has great potential, especially in improving the stability and reliability of power systems, integrating renewable energy protection into the grid.

In the field of vibration signal classification, Jin and Chen [74] proposed a new end-to-end framework consisting of four parts: data preprocessing, time-frequency feature extraction, a modified Transformer network, and overall optimization. In asset maintenance, this framework can significantly improve the efficiency of operational monitoring and maintenance of mechanical structures through real-time monitoring of vibration signals and early fault detection. Time-frequency feature extraction used Mel Frequency Cepstral Coefficients (MFCCs) and Short-Time Fourier Transform (STFT) spectra to enhance the signal feature representation capability. The modified Transformer network utilized a multi-head attention mechanism to capture high-level discriminative features in the time-frequency features. Overall optimization used the Grey Wolf Optimizer (GWO) to jointly optimize key parameters of different parts to improve classification performance. This method can monitor the health status of assets in real-time, promptly detect and prevent potential faults, thus reducing downtime and maintenance costs.

In the area of defect identification in hard disk drives, Kirdponpattara et al. [75] developed a rule-based expert system to optimize the manufacturing process of hard disk drives. This system identifies potentially defective hard disk drives through predictive modeling, classifying them before they undergo exhaustive testing. This approach not only accelerates defect detection but also reduces testing time and resource consumption, thereby enhancing the efficiency of the production process and having significant applications in asset maintenance.

Echraibi et al. [76] developed an expert system based on Deep Infinite Mixture Models for discovering new fault types in Fiber-to-the-Home (FTTH) networks. This system has significant applications in asset maintenance. It combines the strengths of neural networks and infinite Gaussian mixture models by extracting useful features

through the neural network and then clustering the data using these features through the infinite mixture model. This enables the identification of new fault types in unlabeled data.

In the identification of traditional two-lane road types, Barreno et al. [77] developed a fuzzy expert system based on geometric features of the road, such as lane width, shoulder width, slope, and curvature. The expert system employs fuzzy logic to handle the uncertainty and vagueness in the data, generating classifications of road types. This system not only improves the accuracy of road type identification but also has significant applications in asset maintenance.

D. FAULT DIAGNOSIS

The application of expert systems in fault diagnosis can be considered to encompass the whole industrial sphere, whether it is mechanical equipment fault diagnosis, transformer fault diagnosis, or the direction of oil cracking units, all of which have a wide range of applications.

In the fault diagnosis of a pilot-scale distillation column in a DCS environment communication, Leung and Romagnoli [12] developed a fault diagnosis expert system based on a probabilistic model. It provided operators with quick and accurate decision support, helping them make quick and accurate decisions in abnormal situations. In terms of asset maintenance, this system significantly improved the operational efficiency and reliability of equipment through real-time monitoring and precise diagnosis. The authors also emphasized the system's importance in enhancing fault diagnosis efficiency and reducing maintenance costs, particularly in optimizing maintenance strategies and minimizing unplanned downtime, demonstrating the system's efficiency and potential economic benefits.

In the field of fault diagnosis of transformers and high-voltage vacuum circuit breakers, Ma et al. [78] proposed a new multi-layer Back Propagation (BP) network method to solve the problem of reduced convergence speed when the number of network layers increases. Compared with BP Neural Network (BPNN) and BP Expert System (BPES), the results showed that the multi-layer BP network had higher accuracy and speed in fault diagnosis. This method significantly optimized maintenance strategies for critical equipment such as transformers and high-voltage circuit breakers by enhancing diagnostic efficiency and accuracy, thereby improving the system's reliability and stability. The researchers commented on the advancements of this technology, highlighting its importance in increasing maintenance efficiency and reducing operational costs in power systems, and how technological innovation can achieve predictive maintenance and fault prevention.

Similarly, in the fault diagnosis of transformers based on time-frequency analysis, Purkait and Chakravorti [79] proposed a new fault diagnosis expert system that mimicked the performance of experienced testers, using a reasoning engine to perform deductive reasoning based on rules in the

knowledge base and different statistical techniques to identify and locate faults. This system provided necessary information and interaction for non-experts, allowing staff to perform fault diagnosis in a user-friendly window environment. Additionally, the system greatly improved the maintenance and management efficiency of transformer assets through precise fault identification and location. The authors provided a detailed evaluation of this system's contribution to improving fault diagnosis speed and accuracy, noting its crucial role in enhancing power system maintenance efficiency and reducing downtime.

Induction Motors (IMs) can power most modern industrial processes, and continuous monitoring of machine status can detect faults early and avoid costly production process downtime. Machine Current Signature Analysis (MCSA) has become an important technology for condition-based maintenance, but MCSA's industrial application requires trained maintenance personnel who can interpret the current spectrum and avoid misdiagnosis due to electrical noise in harsh industrial environments. Burriel-Valencia et al. [80] proposed a hyper-system implementation to generate an optimized fault diagnosis system to address this issue. This hyper-system has been experimentally tested and verified, confirming that it achieved the proposed objectives. The implementation of this hyper-system not only achieved the proposed objectives but also greatly improved asset maintenance efficiency through real-time accurate monitoring and fault analysis. This system's implementation is significant for improving fault diagnosis accuracy and reducing maintenance costs, especially in asset maintenance management, demonstrating how expert systems can effectively optimize and streamline the fault diagnosis process.

The fluid catalytic cracking unit (FCCU) is one of the most important production processes in the petroleum refining industry, with a complex reaction mechanism that makes it difficult to establish accurate mathematical models. To address this issue, Qian et al. [81] developed a real-time fault diagnosis system for the complex petrochemical process FCCU in 2003. This expert system can quickly diagnose common faults and pinpoint the root cause of the fault, providing operators with suggestions for solving the problem. The system significantly improved asset operational efficiency and reliability through real-time monitoring and diagnosis, effectively preventing potential catastrophic accidents and ensuring equipment safety and production stability.

In 2005, Qian et al. [82] improved this system by introducing a real-time expert system based on wavelet transform and fuzzy ART neural network for fault diagnosis, improving fault detection accuracy. They also improved the user interface to make user interaction smoother. These improvements significantly enhanced the system's capabilities in asset maintenance, particularly in real-time monitoring and fault prediction, helping optimize maintenance strategies and reduce equipment failure rates. The authors highlighted the significant contribution of this technology in improving fault diagnosis efficiency and accuracy in chemical process

operations, emphasizing its important role in maintaining optimal equipment performance and extending equipment life.

In the field of aviation electronics equipment testing, Jie and Shen [83] proposed a knowledge base architecture based on Fault Query Network (FQN), where FQN consisted of Test-Diagnose Fault Units (TDFU). They designed a Fault Answer Best Selection Algorithm (FABSA) with an improved product reasoning engine, extracting the best fault answer based on the alienation degree matrix. This method comprehensively considered shallow knowledge, deep knowledge, and the membership degree of each fault cause. Therefore, it provided comprehensive and accurate support for the fault diagnosis of aviation electronics equipment. This system enhanced the maintenance and reliability of critical aviation assets through accurate fault diagnosis. The authors discussed in detail the application of this method in improving fault diagnosis efficiency, evaluating its contribution to reducing misdiagnosis rates and enhancing maintenance decision quality, emphasizing the system's practicality and the importance of fault prevention.

Similarly, Niccolai et al. [84] proposed an automated detection system focused on the automation of the Fluorescent Penetrant Inspection (FPI) process in aerospace production lines. Their vision-based expert system was designed to detect defects in mechanical parts using image recognition technology. Notably, as illustrated in Figure 5, by applying the t-Distributed Stochastic Neighbor Embedding (t-DSNE) technique to the neural network's classification results, they demonstrated the system efficiency in defect detection, where different types of defects were distinctly separated in a low-dimensional space. This automation not only reduced detection time but also improved the working conditions of inspection personnel. This system significantly improved the maintenance efficiency of production assets by enhancing detection speed and accuracy, ensuring the continuity of production processes and product quality. The authors highlighted the significant contributions of this technology in improving operational efficiency and reducing human error, emphasizing its importance in implementing automation in industrial production.

Lin et al. [51] developed an expert system that enhances the efficiency of software fault localization, which has significant applications in asset maintenance. This expert system integrates various existing Spectrum-Based Fault Localization (SBFL) techniques and uses fuzzy logic to improve the accuracy of locating faulty code. By fuzzifying the outputs generated by different techniques and applying fuzzy reasoning, the system can synthesize more accurate fault suspicion assessments, optimizing the order of code review and improving the efficiency and effectiveness of fault diagnosis.

In the field of object surface detection, Singh et al. [85] integrated deep learning algorithms into an automated vision-based surface defect detection expert system to enhance quality control and asset maintenance in the manufacturing industry. The study highlighted the practicality of

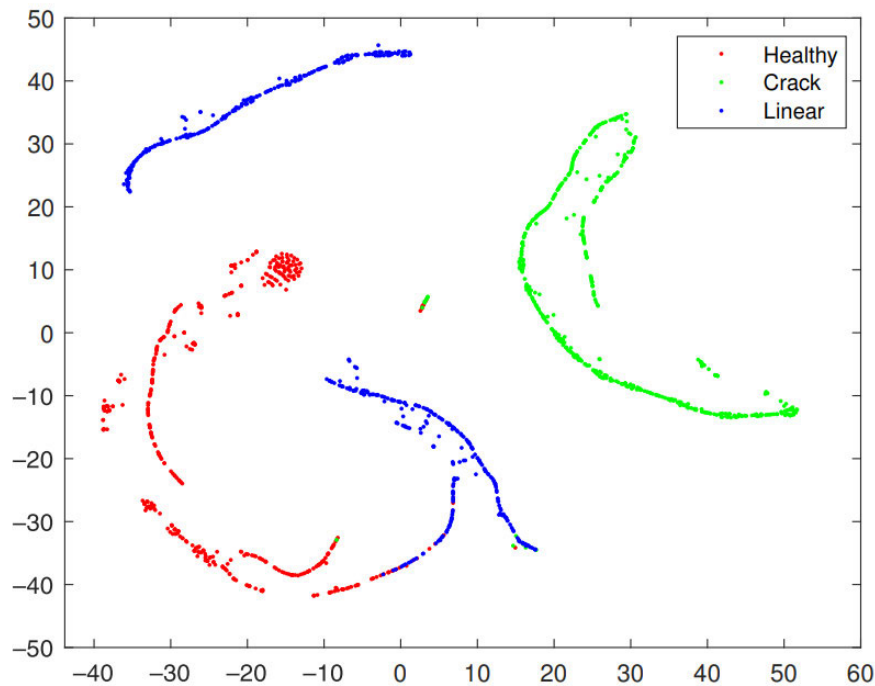


FIGURE 5. t-DSNE visualization of the final classification [84].

pre-trained networks for detecting surface defects in specific mechanical components and emphasized that this approach can significantly improve asset maintenance efficiency for small and medium-sized enterprises (SMEs) and micro, small, and medium enterprises (MSMEs).

In the area of aircraft health management, Ezhilarasu and Jennions [86] developed a system named FAVER (Framework for Aircraft Vehicle Reasoning) to improve the health management of aircraft. This system has significant applications in asset maintenance. FAVER leverages digital twin and reasoning technologies, and by integrating the Open Standard Architecture for Condition-Based Maintenance (OSA-CBM) framework, it can monitor and diagnose faults at the entire vehicle level, particularly those faults that propagate across different systems.

E. OTHER APPLICATIONS

Expert systems contribute to improving efficiency, enhancing security, and reducing costs in both modern industries and blockchain fields. They can provide intelligent decision support and automate a series of tasks, making businesses more competitive.

To provide intelligent services for IoT layout design, Carreño et al. [87] proposed a system called the Blockchain-based IoT Expert System Shell (BMES). BMES combined the concept of Initial Coin Offerings (ICOs) with Expert System Services (ESS) that leverage expert knowledge as a form of cryptocurrency trading. The ESS employs expert knowledge to develop intelligent asset maintenance strategies, ensuring enhanced system operational continuity.

Here, the value of ESS depends on its demand and benefits, ensuring data security and reliability through ICO transactions and verification processes based on blockchain technology. The authors believed that this research combined two cutting-edge technologies: blockchain and artificial intelligence, providing an innovative solution for customized production and the new industrial revolution, by applying expert knowledge to improve the efficiency and credibility of production and IoT systems.

To better optimize sensor distribution and control in smart cities, Aguilera et al. [88] proposed an IoT expert system based on blockchain technology, using the EOSIO platform as the blockchain foundation, employing smart contracts and neural networks as reasoning engines. This expert system leverages expert knowledge to provide intelligent recommendations and decision-making. Based on users' needs and preferences, it provided recommendations on the quantity and location of sensors for different spaces (parks, schools, hospitals, etc.) and took corresponding measures (such as adjusting lighting, activating fire alarms, temperature alarms, etc.) when sensor readings exceeded normal ranges. The integration of expert knowledge ensures that the recommendations are accurate and tailored to specific scenarios, enhancing the system's effectiveness and reliability.

Moreover, the expert system saves sensor coordinate data on EOS nodes, utilizing the security and decentralization features of blockchain, while other data are saved in the cloud, achieving data management and semantic search in the IoT environment.

TABLE 6. The scenarios, optimization algorithms, application of AI/ML technologies in asset management and other aspects of application cases.

Scenario	Optimization Algorithm	AI/ML Application in Asset Management	Other Important Points
Design, Manufacturing, Maintenance, Recycling/Retrofitting	Genetic Algorithms (GA), Convolutional Neural Networks (CNN), Support Vector Machines (SVM)	Applied in lifecycle phases to optimize process control, improve productivity and quality	Supports Industry 4.0, challenges with legacy systems, data privacy, benefits include enhanced decision-making, cost reduction
Condition monitoring, Health index, Lifecycle enhancement, Fault diagnosis	Dissolved Gas Analysis (DGA), Furan Analysis, Polarization and Depolarization Current (PDC) Analysis	Used for diagnosing faults, predicting lifespan, and enhancing condition-based assessment	Challenges include technical complexities, data privacy issues, and integration with existing systems. Benefits include improved reliability, cost-effectiveness, and proactive maintenance strategies.
Condition monitoring, Fault diagnosis, Health index, Insulation status, Lifecycle enhancement	Fuzzy Logic Controller (FLC), Fuzzy Clustering Means (FCM), Genetic Algorithms (GA)	Diagnosis and prognosis using dissolved gas analysis (DGA), Furan Analysis, Interfacial Tension (IFT), Degree of Polymerization (DP)	Enhances decision-making, improves reliability, cost-effectiveness, proactive maintenance, supports Industry 4.0 initiatives, addresses technical complexities, and data privacy issues
Risk ranking of natural gas pipelines	Multivariable Predictive Control (MPC)	Real-time prediction and assessment of pipeline reliability and risk	Transition from "after-failure" to preventative maintenance based on predictive reliability data
Reliability monitoring and predictive control of oil and gas facilities	Multivariable Predictive Control (MPC)	Real-time prediction and assessment of facility reliability based on multiple factors	Transitions from "after-failure" maintenance to preventative maintenance using predictive reliability data
Safety performance of Malaysian offshore oil platforms	Statistical correlation analysis using Pearson correlation	Evaluating and predicting safety performance through key safety factors	Integrative safety performance measurement combining leading and lagging indicators for proactive and reactive monitoring
Drug-excipient compatibility evaluation	Rule-based expert system	Predicts incompatibility risks based on a comprehensive database of known interactions	Integrates an extensive database with 532 data items from 228 articles; provides detailed compatibility evaluation with user-friendly interface
Cold supply chain risk evaluation	Entropy-Ordered Weighted Average (OWA) method	Evaluates and ranks risks using expert system knowledge base	Integrates subjective expert knowledge with objective data to reduce bias; applied to fresh food e-commerce logistics
Biomass polygeneration system in Malaysia	Decision and Evaluation-based Fuzzy Analytic Network Process (DEFANP)	Prioritizing risk mitigation strategies for decision-making purposes	Monte Carlo simulation used for modeling fluctuations and uncertainties
Supplier selection in the defense industry	q-Rung Orthopair Fuzzy Set (q-ROFS) and EDAS	Uses fuzzy-based decision-making models for evaluating supplier performance	Incorporates Delphi method and AHP to determine criteria; combines subjective expert knowledge with objective data for better decision-making
Improving recommendation accuracy in collaborative filtering systems	Expert mining system using signal processing and kurtosis factor	Enhances credibility of expert recommendations by filtering based on expertise level	Uses signal fluctuation and kurtosis factor to determine expert credibility; tested on multiple datasets (MovieLens, Jester, Booking-Crossings, Last.fm) showing improved accuracy
The article discusses a framework for improving fault localization in software debugging.	Fuzzy Expert System (FES)	Integrates multiple Spectrum-Based Fault Localization (SBFL) techniques to enhance fault localization effectiveness.	The framework does not require historical data for model training, making it suitable for new software projects and complements existing approaches .
Prioritizing industries for installing combined heat and power (CHP) systems in Iran	Analytical Hierarchy Process (AHP) and Fuzzy Expert System (FES)	Decision-making model integrates AHP and FES to rank industries for CHP installation	Focuses on reducing energy costs and improving efficiency in various industries; validated against developed countries' CHP usage patterns
Selection of environmentally friendly composite materials for automotive components	Analytical Hierarchy Process (AHP)	Expert system aids in selecting optimal materials considering environmental and sustainability factors	Prioritizes environmental factors; hemp and polypropylene (PP) ranked highest for automotive components
Mechanical failure diagnosis in cars	Case-based Reasoning (CBR)	Utilizes past cases to diagnose and recommend	The system is designed to be an online application for use by both technicians and

TABLE 6. (Continued.) The scenarios, optimization algorithms, application of AI/ML technologies in asset management and other aspects of application cases.

		solutions for mechanical issues in cars	general users; allows for easy updates and knowledge improvement by authorized experts
Steel production planning and scheduling	Shuffling-aided network method	Assessing complexity of selection patterns established by human experts	Decision-support systems rely on tacit knowledge for steel grade selection based on carbon content, while other elements play a lesser role
Fault localization in software debugging	Fuzzy Expert System (FES)	Integrates multiple SBFL techniques	Combines outputs of multiple techniques to enhance fault localization effectiveness
Selection of wastewater treatment plants	Pythagorean Fuzzy Hypersoft Matrices (PFHSMs)	Improves decision-making accuracy by considering multiple criteria	Combines the advantages of Pythagorean fuzzy sets and hypersoft sets for flexible and reliable decision-making; used in real-world applications like mobile selection and wastewater treatment plant selection
Fault discovery in complex systems	Deep Infinite Mixture Models	Improved fault localization and prediction accuracy	Use of a Bayesian non-parametric approach to handle an infinite number of mixture components
Object classification using ultrasonic sensors in industrial applications	Fuzzy Expert System with dual knowledge base (statistical and rule-based)	Real-time adaptation to environmental changes, such as temperature and humidity, for reliable classification	Automatically generates and updates knowledge base and decision rules; uses Jeffreys-Matusita criteria for feature selection to improve accuracy
Classification of welding flaw types	Fuzzy Rule-Building Expert System (FuRES)	Automated identification of welding defects in radiographic images	Outperforms fuzzy K-nearest neighbors and multi-layer perceptron neural networks in classification accuracy
Classification of various fuel types (e.g., rocket, diesel, jet fuels)	Fuzzy Rule-Building Expert System (FuRES)	Accurate classification of fuel types based on GC/DMS data with a 95% success rate	Robust model demonstrated by accurate classification of fuel samples after one month; utilizes principal component transform for data compression
Classification of jet fuels by lot numbers	Fuzzy Rule-Building Expert System (FuRES)	Real-time classification of jet fuels based on GC/MS data	Achieved 99.8% classification accuracy; robustness validated with multiple data sets and time-separated samples
Online prediction of iron ore grades	Sequential Forward Floating Selection (SFFS) with SVR	Machine vision system for real-time quality monitoring of iron ore	High accuracy in grade prediction using optimised feature subset and SVR; superior performance compared to other models like GPR and ANN
The use of an expert system to automate security testing in IoT ecosystems	ESSecA leverages a multi-phase methodology for automated threat model generation and attack plan creation	Utilizes threat intelligence sources like CWE, CAPEC, and ATT&CK for threat modeling and penetration testing	The system integrates various threat intelligence data sources and produces outputs such as Threat Models and Attack Plans to enhance security posture assessment
Classification of power quality disturbances in power systems	S-transform integrated with a multi-layer perceptron neural network using resilient backpropagation (RPROP)	Automated classification of power quality disturbances based on extracted features from S-transform	Effectively classifies multiple types of disturbances, including voltage sags, swells, and harmonics, under different noise conditions
Classification of power quality events during the integration of renewable energy sources	Analytic Hierarchy Process (AHP)	Uses linear discrimination analysis for feature extraction and GMDH for classification	Real-time validation with sample data confirms high classification accuracy for various power quality disturbances
Health monitoring of a rotating machinery under constant loading conditions	Grey Wolf Optimizer	Transformer network modified for high-level feature extraction from vibration signals	Multi-sensor fusion enhances classification accuracy compared to single-sensor approaches
Hard disk drive manufacturing optimization	Decision Tree (C4.5 with Information Gain)	Predicts defective drives to reduce testing time and resource consumption	Achieves high prediction accuracy and reduces total test time significantly by preemptively identifying defective drives
Fault diagnosis in GPON-FTTH networks	Deep Infinite Mixture Models	Identifies and classifies new types of network faults without needing prior labels	Combines neural networks with Gaussian mixture models for clustering; capable of handling high-dimensional data and discovering zero-day anomalies

TABLE 6. (Continued.) The scenarios, optimization algorithms, application of AI/ML technologies in asset management and other aspects of application cases.

Identification and risk assessment of two-lane roads	Mamdani fuzzy system and ANFIS	Classifies road types and assesses driving risks based on road geometry	Improves road safety by updating classifications and recommending speed limits; identifies high-risk sections for targeted interventions
Fault diagnosis in a chemical plant's distillation column	Dynamic Causal Network Management (DCNM)	Bayesian belief network for probability calculations	Effective handling of phantom alarms caused by process time-delays
Energy efficiency diagnostics in Brazilian industrial plants	Fuzzy AHP-TOPSIS	Knowledge management for consistent and reliable solutions	Provides detailed energy efficiency recommendations and explanations automatically
Impulse fault diagnosis in power transformers	Re-confirmative approach combining time-domain and frequency-domain analyses	Expert system provides reliable fault detection and location determination for transformer winding insulation failures	Integrates heuristic rules with statistical methods; tested on 3, 5, and 7 MVA transformers with high accuracy and reliability
Fault diagnosis for induction motors under transient regime	Harmonic Order Tracking Analysis (HOTA)	Use of support vector machine (SVM) and artificial neural network (ANN) expert systems for fault classification	The supra-system automates the generation and optimization of diagnostic systems, achieving high diagnostic accuracy and efficiency
Real-time fault diagnosis in chemical processes	Heuristic bounded deep first search strategy	Expert system provides real-time decision support for fault diagnosis	Knowledge base uses multi-level structure for efficient rule access
Fault diagnosis for fluid catalytic cracking unit (FCCU)	Integrity verification algorithms for knowledge base maintenance	Enhances reliability and efficiency of fault diagnosis system by maintaining consistent and accurate knowledge base	Utilizes DELPHI and SQL SERVER for development; ensures safety and stability in FCCU operations
Non-destructive testing (NDT) for detecting cracks and defects in aerospace components	Feature extraction combined with neural networks (direct and indirect classifiers)	Automated detection and classification of defects in mechanical parts using image processing	Achieved high classification accuracy with limited training samples; improves reliability and reduces human operator workload
Failure mode and effects analysis (FMEA) for risk evaluation in engineering applications	Interval Area Metric (IAM) and Dimensionless Uncertainty Metric (DUM)	Expert system qualitatively evaluates uncertainty metrics for better decision-making in asset management	Improves risk assessment accuracy and helps in identifying the most critical failure modes under uncertainty
Assessing energy consumption of electric cars in various driving cycles	Multi-Valued Logic Trees	Provides decision support for energy management and driving style adjustments	Integrates ambient temperature and driving speed data to optimize energy consumption; offers actionable prompts to drivers for improved efficiency
Fault diagnosis and health state evaluation of lithium battery pole double rolling equipment	Fuzzy Expert Fault Diagnosis Model (FEFDM) and BP Neural Network	Combines fuzzy logic with BP Neural Network to diagnose faults and assess equipment health	Integrates Extended Kalman Filtering (EKF) to clean data and filter out noise, improving diagnostic accuracy
IoT installation layout designs using blockchain technology	Extreme Learning Machine (ELM)	Designing customized IoT installation layouts based on user inputs	Smart contracts ensure secure transactions; decentralization enhances data integrity and protection against cyber-attacks
Smart cities: optimizing sensor distribution and actions based on sensor data	Deep Neural Network (DNN) with Kalman filters	Recommends sensor distribution and takes automated actions (e.g., control lights, activate alarms) based on sensor readings	Uses EOSIO blockchain for secure, decentralized data management and smart contracts
The use of an expert system to automate security testing in IoT ecosystems	ESSecA leverages a multi-phase methodology for automated threat model generation and attack plan creation	Utilizes threat intelligence sources like CWE, CAPEC, and ATT&CK for threat modeling and penetration testing	The system integrates various threat intelligence data sources and produces outputs such as Threat Models and Attack Plans to enhance security posture assessment
Real-time safety monitoring of hydropower dams	Sequence monitoring using Wavelet analysis	Expert system for identifying data anomalies and evaluating dam safety	Applied to over ten dam projects in China, improving efficiency and decision-making

TABLE 6. (Continued.) The scenarios, optimization algorithms, application of AI/ML technologies in asset management and other aspects of application cases.

Preliminary diagnostics of energy efficiency in Brazilian industrial plants	Rule-based expert system	Provides recommendations for energy efficiency improvements based on expert knowledge	Knowledge management in SMEs, creating permanent corporate memory for energy efficiency expertise
Operating regime detection in industrial gas turbines	Heuristic bounded deep first search strategy	Real-time detection of start-up, stand-still, turn-off regimes based on sensor data	Reduces computational complexity and memory usage for efficient real-time performance
Optimization of laser welding for additively manufactured thermoplastic parts	Neural Network-based expert system	Predicting shear tensile force to improve weld seam quality	Achieved 88.1% accuracy in predicting weld strength; uses multiple manufacturing and welding parameters to optimize outcomes

In the security assessment process of IoT devices and IoT infrastructure, Rak et al. [89] developed an expert system called ES_{SecA} to support security experts and penetration testers in automated threat modeling and penetration testing plan generation in the above process. ES_{SecA} leveraged expert knowledge and utilized multiple threat intelligence knowledge bases to analyze potential risks and vulnerabilities of the target system. By incorporating expert knowledge, the system provided precise and actionable insights to enhance the security assessment process. The authors demonstrated the effectiveness and practicality of ES_{SecA} by presenting a typical home automation system, the Open Energy Monitoring Platform.

In the field of water dams, developing high-performance safety monitoring systems plays an important auxiliary role in dam safety management work. Based on this, Han et al. [90] developed an expert system for dam safety management, which leverages expert knowledge for real-time monitoring of dam safety throughout the process, covering abnormal situations. The system incorporates expert insights to provide rich functions such as dam safety analysis, visual query, remote consultation, etc. By utilizing expert knowledge, the system enhances the accuracy and reliability of safety assessments and decision-making processes in dam safety management.

In a study on the preliminary diagnosis of energy efficiency (EE) potential and knowledge management (KM) for Brazilian industrial plants, Bucciari et al. [91] developed a prototype expert system (ES) that utilized the expertise and experience of a group of EE experts and industrial sector professionals. This expert knowledge was represented in the form of rules and implemented through an open-source tool CLIPS. By leveraging expert knowledge, the prototype also contains an explanation module to support decision-making and provide insights on taking EE actions.

To monitor the operating status of industrial gas turbines, including startup, shutdown, steady state, load, and the activated controller type, such as the startup controller and speed controller, Nayeri et al. [92] developed a rule based expert system that leverages expert knowledge. Based on the control system logic of gas turbines, technicians’ knowledge, and

actual data, a set of expert rules were designed to judge the operating status based on sensor data such as shaft speed, temperature, pressure, etc. By incorporating expert knowledge, the authors optimized the complexity and number of parameters of the rules to embed the expert system into embedded hardware and evaluated it using data from different manufacturers and environmental conditions.

Differently, in predicting shear tensile force, Kuklik et al. [93] proposed an artificial neural network method, using polylactic acid as the material, and made transparent and black samples. Then, laser welding was performed with different laser powers and scanning speeds, followed by shear tensile tests on the welded samples, and the transmittance of transparent samples was measured with a spectrometer. The authors used 320 experimental data to train and validate the artificial neural network to evaluate its prediction accuracy and stability. In contrast, the expert system method leverages expert knowledge to predict outcomes by incorporating domain-specific insights and rules. The results showed that the artificial neural network could predict shear tensile force with an accuracy of 88.1%, an expert system utilizing expert knowledge could potentially offer enhanced interpretability and reliability in predictions.

In many industrial production processes, high levels of expertise and complex analysis capabilities are required, and expert systems can meet these production needs by simulating human experts’ decision-making and reasoning processes. Therefore, the widespread application of expert systems in industrial production not only improves efficiency and accuracy but also provides more reliable tools for decision-makers to meet the challenges of a constantly changing production environment. Risk assessment, decision analysis, industrial classification, and fault diagnosis are all critical components of industrial production, and expert systems are employed to facilitate these procedures. It shows the wide application and important role of expert systems in the industrial field.

The Table 6 summarizes typical application cases of expert systems in the industrial field, including application scenarios, optimization algorithms used, applications of AI/ML technologies in asset management, and other important aspects. The table provides a detailed display of the

operational processes, algorithms and tools used, data processing methods, and other technical content in different application scenarios. It also conducts an in-depth analysis of the performance and effects of these technologies in practical applications. These case studies not only demonstrate the superiority and practicality of various technologies in real-world applications but also reveal their potential in different industrial environments. Through the analysis of these cases, readers are provided with a comprehensive technical perspective and key insights. Furthermore, the table emphasizes the important role of expert systems in improving production efficiency, reducing costs, ensuring product quality, and enhancing safety. These technical contents and key comments not only verify the effectiveness of various technologies in practical applications but also highlight directions for future improvement and optimization, providing valuable references for researchers and practitioners.

V. DISCUSSION

In previous studies, we have analyzed numerous instances of expert systems applied in the industrial sector. These instances demonstrate that expert systems can significantly improve production efficiency, enhance product quality, and reduce production costs. By utilizing rule-based and data-driven reasoning mechanisms, expert systems can greatly enhance the automation and intelligence level of production processes, minimize human errors, and improve operational efficiency and safety. The importance of these findings lies not only in validating the practical feasibility of expert systems in addressing complex industrial problems but also in providing empirical foundations for designing, implementing, and optimizing these systems.

Despite the significant advancements in the application of expert systems in the industrial field, there are still some research gaps evident from the current literature and practical applications.

A. INTEGRATION WITH ADVANCED AI TECHNOLOGIES

Currently, the integration of expert systems with advanced artificial intelligence technologies has not been fully developed, limiting their potential application in broader scenarios [94]. In the future, with the rapid advancement of AI technologies such as deep learning and machine learning, integrating these technologies with existing expert systems will open up many new possibilities. For instance, Mewada et al. [98] developed a machine learning-based intelligent diagnostic system that effectively identifies and diagnoses defects in the forging process, offering new possibilities for automated and intelligent diagnostics. Asif et al. [43] used a deep learning expert system to automatically process transient electromagnetic (TEM) data from ground and aerial systems, significantly improving the efficiency and accuracy of data processing.

B. INTERPRETABILITY AND USER-FRIENDLINESS

With the rapid development of deep learning, neural network-based expert systems have shown promising

decision support capabilities. However, due to the “black box” nature of neural networks, the interpretability and user-friendliness of these expert systems need improvement [95], [97]. The transparency of the decision-making process directly affects users’ trust in the system’s outputs, which is particularly crucial for the acceptance and trust of non-professional users [96]. Therefore, increasing the interpretability of expert systems, especially those based on deep learning, by introducing interpretable models and algorithms, can help users understand the system’s decision-making process, thereby increasing trust. For example, Cao et al. [99] proposed a novel hierarchical belief rule-based (H-BRB) expert system that improves model interpretability while maintaining accuracy.

C. CROSS-DOMAIN APPLICATION AND COLLABORATION

The case analyses indicate that existing expert systems are often focused on applications within a specific domain. There is a need for further research and development to enhance their cross-domain generality and flexibility to meet the rapidly changing industrial demands. In the future, increasing the cross-domain generality of expert systems and strengthening collaboration with experts from other disciplines through interdisciplinary knowledge integration will be a major direction for future efforts. Suzumura et al. [100] pointed out that interdisciplinary collaboration and the use of cloud platforms such as mdx can support research in different fields through high-performance computing and secure data sharing. These technologies promote knowledge sharing and collaboration, driving the application and innovation of expert systems in a broader range of fields, developing more comprehensive and efficient expert systems.

In summary, expert systems have demonstrated great potential in the industrial sector, but further research and development are needed in integrating advanced AI technologies, improving interpretability and user-friendliness, and enhancing cross-domain generality and collaboration. By advancing research and practice in these areas, it is possible to develop more intelligent, efficient, and comprehensive expert systems to meet the complex and ever-changing demands of future industries.

VI. CONCLUSION

In this study, the application of expert systems in the industrial field has been comprehensively reviewed. Initially, the fundamental concepts of expert systems were detailed (Section II), including their components such as knowledge bases, inference engines, user interfaces, interpreters, and knowledge acquisition systems, as well as expert systems based on different reasoning mechanisms like rules, cases, and neural networks (Section II-A), along with the developmental history of expert systems encompassing four generations (Section II-B). Subsequently, the advantages (such as improved production efficiency and product quality) (Section III-A) and limitations (such as dependency on expert knowledge and potential errors) (Section III-B) of expert

systems were explored. Finally, application cases of expert systems across various segments of industrial production were discussed and analyzed.

When discussing and analyzing application cases of expert systems in various segments of industrial production, it was found that expert systems often outperform other methods in solving problems in traditional industrial fields [46], [139], including areas such as risk assessment, decision analysis, industrial classification, and fault diagnosis (Section IV). Additionally, a key conclusion was revealed through analysis: the optimal efficacy of expert systems primarily depends on the integration of a comprehensive and accurate expert knowledge system and reasoning rules, along with the comprehensiveness and accuracy of input data.

However, expert systems also face challenges in industrial applications. The current expert systems are not fully developed in integrating with emerging technologies. With the application of deep learning technologies to expert systems, while enhancing the decision-making capabilities, the interpretability of expert systems has decreased [76], [140]. Moreover, our analysis of numerous cases indicates that the expert systems developed by researchers are mostly applied in specific fields, and further development is needed for cross-domain applications.

Looking to the future, despite facing a range of challenges, expert systems continue to demonstrate significant potential and value in the industrial field. To further advance their development, it is crucial to deeply integrate expert systems with advanced artificial intelligence technologies such as deep learning and machine learning. Additionally, improving the interpretability of the systems by introducing transparent models and algorithms that allow users to clearly understand the decision-making process is also an important direction. Finally, through interdisciplinary collaboration, integrating knowledge from multiple fields to develop more comprehensive and efficient expert systems will bring more innovation and breakthroughs to industrial applications. In conclusion, future expert systems will continue to play a significant role in driving intelligent upgrades in the industrial field, propelled by technological advancements and cross-domain cooperation.

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